

Zaroor boss! Yeh lijiye, aapka Waste Classification Prototype ka material bhi bilkul MS Word aur PDF ke liye clean aur ready kar diya hai. Saare extra symbols (jaise stars aur backticks) hata diye hain taake aap directly copy-paste kar sakein.

Yahan se copy karein:

WASTE CLASSIFICATION PROTOTYPE - VIVA PREPARATION NOTES

PROJECT DETAILS

Project Title: Waste Classification Prototype using Flask and TensorFlow/Keras

Model: MobileNetV2 Transfer Learning

Team/Student: _____

Course: _____

Date: _____

SECTION 1: SHORT VIVA NOTES

1. PROJECT OVERVIEW

- Project ka goal waste image ko classify karna hai into classes (jaise Plastic, Metal, Glass, Biodegradable, Paper).
- System user ko image upload ya capture karne deta hai aur prediction ke sath confidence score show karta hai.
- Backend Flask par banaya gaya hai, aur model TensorFlow/Keras transfer learning (MobileNetV2) use karta hai.

2. WHY TENSORFLOW/KERAS?

- Fast prototyping ke liye Keras ki API bohot simple hai.
- Pretrained models easily available hote hain.
- Training, evaluation, aur inference ek hi ecosystem mein smoothly integrate ho jate hain.

3. WHY MOBILENETV2?

- Yeh ek lightweight aur efficient model hai.

- Prototype mein kam resources (CPU/RAM) par bhi achi performance deta hai.
- Transfer learning se kam data par bhi better accuracy milti hai.

4. TRANSFER LEARNING KYA HAI?

- Pretrained model (jo ImageNet par train hua ho) ka feature extractor use kiya jata hai.
- Base layers ko freeze kar diya jata hai, aur upar custom classification head laga kar train kiya jata hai.
- Is se training fast hoti hai aur performance automatically improve hoti hai.

5. DATASET HANDLING

- Dataset class-wise folders mein organize kiya gaya hai.
- Stratified 70/30 split kiya gaya hai (70% train, 30% test), taake har class ki distribution maintain rahe.
- Image validation aur supported extensions filtering apply ki gayi hai.

6. TRAINING STRATEGY

- Training set par data augmentation (rotation, shift, zoom, flip) use ki gayi hai taake model diverse data seekhe.
- Callbacks use kiye gaye hain:
 - EarlyStopping: Overfitting reduce karne ke liye.
 - ModelCheckpoint: Best model save karne ke liye.
 - ReduceLROnPlateau: Learning rate ko stabilize karne ke liye.
- Files jo save hoti hain: best_model.keras, final_model.keras, class_indices.json, aur training_curves.png.

7. EVALUATION METRICS

- Model ko Test set (30% unseen data) par evaluate kiya gaya hai.
- Metrics check kiye gaye: Accuracy, Precision (macro), Recall (macro), aur F1-score (macro).
- Confusion matrix generate karke confusion_matrix.png mein save ki gayi hai.

8. ACCURACY vs PRECISION vs RECALL vs F1

- Accuracy: Total correct predictions ka ratio.
- Precision: Jo positive predict kiye, un mein se actual kitne sahi the.
- Recall: Total actual positives mein se model ne kitne identify kiye.
- F1-score: Precision aur recall ka balanced harmonic mean hai.

9. CONFUSION MATRIX KA PURPOSE

- Har class ke true labels versus predicted counts dikhati hai.

- Is se pata chalta hai ke model kin classes ke beech confuse ho raha hai (misclassification pattern).
- Class-wise weakness detect karne mein bohot helpful hai.

10. FLASK APP FLOW

- GET / route: User Interface (UI) load hoti hai.
- POST /predict route: Image upload ya captured data receive hota hai, aur model prediction return karta hai.
- GET /metrics route: Latest metrics aur confusion matrix ki info return karta hai.

11. UI FEATURES

- Upload image input aur browser camera capture support.
- Prediction class, confidence score, aur class probabilities display hoti hain.
- Metrics dashboard jisme Accuracy, Precision, Recall, F1 aur confusion matrix شامل hai.

12. CLASS_INDICES.JSON KYUN IMPORTANT HAI?

- Training ke time class names aur unke index ki mapping save hoti hai.
- Inference (prediction) ke time yahi same mapping use hoti hai.
- Is se label mismatch ki problem avoid hoti hai.

13. ERROR HANDLING (IMPLEMENTED)

- Agar model ya artifacts missing hon to error handle hota hai.
- Invalid image upload par warning aati hai.
- Camera permission denied ya unavailable hone par fallback messaging di gayi hai.

14. PROJECT LIMITATIONS

- Yeh ek prototype level system hai (production level security aur auth abhi nahi hai).
- Real-time video stream inference support nahi karta, sirf images par kaam karta hai.
- Dataset ki quality aur class imbalance model ki performance ko directly impact karti hai.

15. FUTURE IMPROVEMENTS

- Fine-tuning stage (kuch base layers ko unfreeze kar ke dobara train karna).
- Class imbalance ko handle karne ke liye class weights use karna.
- Docker deployment aur cloud hosting.

SECTION 2: 10 MOST EXPECTED VIVA QUESTIONS

Q1: Aap ne kaun sa model use kiya? Answer: Hum ne MobileNetV2 use kiya hai transfer learning ke sath, jisme custom dense head lagaya gaya hai.

Q2: Transfer learning kyun use ki? Answer: Kyunke yeh limited data par faster training aur better performance deta hai.

Q3: 70/30 split kyun kiya? Answer: Taake 70% data par achi training ho sake aur 30% unseen data par balanced evaluation ho sake.

Q4: Stratified split ka kya benefit hai? Answer: Is se har class ka proportion training aur testing data dono mein consistent rehta hai.

Q5: Accuracy aur F1-Score mein kya farq hai? Answer: Accuracy overall correctness batati hai, jabke F1-Score precision aur recall ka balance batata hai (khaas taur par unbalanced data mein).

Q6: Confusion matrix ka kya use hai? Answer: Yeh class-wise misclassification analysis dikhati hai (konsi class doosri class se mix ho rahi hai).

Q7: Best model kaise save hota hai? Answer: ModelCheckpoint callback ke zariye jo validation loss ko monitor karta hai aur lowest loss wale model ko save karta hai.

Q8: Inference ke waqt label mismatch kaise avoid kiya? Answer: Training ke waqt class_indices.json file save kar ke, taake prediction ke waqt classes same order mein rahen.

Q9: UI se prediction ka flow kya hai? Answer: User image upload karta hai -> request /predict API par jati hai -> model classify karta hai -> class aur confidence wapis UI par show hota hai.

Q10: Prototype completion ka criterion kya tha? Answer: Training, evaluation, artifacts generation, UI prediction, aur metrics display ka successfully kaam karna.